NetC

NetC is a domain-specific language designed with a focus on **logic-flow** and **network-like structures**. It follows a structured, C++ like syntax but uses custom keywords to represent input, output, and specialized functions.

Keyword Reference

The following keywords are reserved in NetC. Using these as variable names will trigger a **Lexical Error**.

Structure & Execution

Keyword	Description
link	Imports external NetC libraries or files.
init	The **entry point** of the program (similar to main in C++).
network	Defines a **function** or a logic block.
yield	Returns a value from a network function.

Data Types (The "Primitives")

Keyword	Logical Type	C++ Equivalent	Memory Logic
dnum	**Discrete Number**	int	Integer values for indexing, counting, and logic.
cnum	**Continuous Number**	float / double	Real numbers for calculations and weights.
text	**String Literal**	std::string	Sequence of characters for messages or labels.
flag	**Boolean Logic**	bool	Binary state: true or false.

Control Flow

Keyword	Description
if / else	Standard conditional branching.
iterate	A fixed-loop structure (similar to for).
until	A conditional loop (similar to while).

I/O Operations

Keyword	Description
feed	Input: Reads data into a variable.
forward	Output: Sends/Prints data to the console.

A Simple Example

```
link "math.ai";

network add(dnum a, dnum b)

{

   yield a + b;

}

init()

{

   dnum x = 5;

   dnum y = 10;

   dnum result = add(x, y);

   forward(result);

}
```